

Mechanistic Interpretability for Latent Truthfulness Detection in Lightweight Transformers

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Abstract—Most modern AI systems generate fluent language but sometimes produce false statements. These statements are sometimes genuine mistakes, but sometimes lies are crafted against the original programming protocols to either ‘trick’ the user or support them. Most current approaches to detection rely on training an outside classifier on outputs to detect lies. Therefore, this study questioned whether ‘neural networks represent truth and falsehood between its layers’ and whether we could use methods to effectively produce a latent truthfulness detection system for AI models.

Index Terms—AI, Mechanistic Interpretability, Latent Truthfulness, Transformers

I. INTRODUCTION

Mechanistic interpretability is a ML safety technical solution which effectively reverse engineers an AI model to decode what stimulates the decisions made by them. It turns the otherwise black box of AI into a white box, and the thinking process could be extracted for helpful research. This has many uses in modern day research, including ensuring AI misalignment and potentially fixing errors that are inbuilt into the network. It relies on a theory that the decisions made by AI are encoded and therefore de-codable.

Recent work in mechanistic interpretability suggests that many semantic features in neural networks are actually hidden as geometric structures within the neuron relations. The Linear Representation Hypothesis, an idea that high-level concepts are represented linearly as directions in some representation space, can be used in reference to the workings of artificial intelligence. Use of this theory with respect to AI also brings a follow on theory - whether a ‘lie detector’ can be created with an mechanistic interpretability, effectively analysing neuron relations to pick up on deception. This has many benefits, predominantly less AI safety risks with lies. The reason a latent truthfulness detection was used was because there are no ‘polygraph tests’ for robots, like there are of humans. Instead, looking at internal structures could be compared to looking at the brain to notice the differences when a lie versus a truth is told.

A. Prerequisites

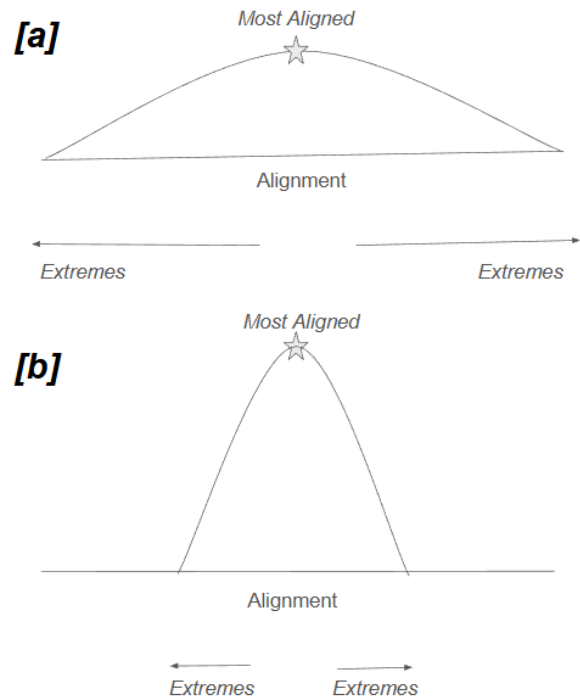


Fig. 1.

a) General ML model alignment diagram. There are two different extremes and an optimal alignment point.

b) ML model with technical safety solutions applied (scalable oversight, mechanistic interpretability). Has shorter extremes and steeper sides.

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The main prerequisite to understanding further terminology is the alignment graph. Essentially, a normal ML graph has general extremes when it is ‘not aligned’ with different goals to humans and may act erratically and lie. The main aim of mechanistic interpretability and many other AI safety solutions is to shorten extremes and ensure misalignment by a big degree has a very small to no chance of occurring. My investigation on whether mech-interp is truly a good solution is whether I believe it actively stretches the graph from Fig.1(a) to Fig.1(b).

B. Paper overview

To explore this question, I trained a small Transformer language model using pytorch on a dataset containing both true and false factual statements. Importantly though, the model was trained only on next-token prediction and was not explicitly instructed to classify statements as true or false. After training, I extracted hidden state representations for each sentence and computed a latent “truth direction” by measuring the difference between the average internal representations of truthful and false statements.

The reason behind this is that if the truthfulness, is truly geometrically encoded, then projecting the sentence (representation) onto this direction should separate truthful from false statements properly. The insightful mini project would prove whether mechanistic interpretability was a viable solution, especially as model complexity rapidly increases, exceeding Moore’s Law by a significant difference. In fact, AI training compute has been doubling every 3.4 to 6 months.

II. METHODS

Machine Learning

A. Rationale

To conduct a mechanistic interpretability analysis on a simple transformer to prove for certain whether there is a way to identify latent truthfulness, and that the decisions behind ML models can be decoded from neuron activations.

The architecture consisted:

- Token embedding layer mapping characters to dense vectors
- Positional embedding layer encoding token order
- Multiple transformer encoder blocks utilising multi-head self-attention
- Final linear projection layer mapping hidden states to vocabulary logits.

Training was performed with cross entropy loss between predicted and target next tokens and optimization was conducted using AdamW with a fixed learning rate. Hidden states for retained for analysis(from the final transformer layer)

B. Sentence Representation Extraction

For each of the innput statements the hidden states (corresponding to all tokens) were extracted from the final transformer layer and to obtain the single vector rep for each sentence token level hidden states were averaged using:

$$h_{\text{sentence}} = \frac{1}{T} \sum_{t=1}^T h_t \quad (1)$$

C. Latent Truth Direction Computation

To identify whether truthfulness was encoded geometrically(the main investigation) mean hidden representations were computed separately for true and false statements.

$$\begin{matrix} \mu_{\text{truth}} \\ \mu_{\text{lie}} \end{matrix} \quad (2)$$

A latent truth direction was then defined as the normalised difference, allowing visualisation:

$$d_{\text{truth}} = \frac{\mu_{\text{truth}} - \mu_{\text{lie}}}{\|\mu_{\text{truth}} - \mu_{\text{lie}}\|} \quad (3)$$

The direction therefore represents the primary axis separating true from false representations in the hidden space.

D. Mechanistic Lie Detection

For any statement x a truth score was computed by projecting the sentence embedding onto truth direction:

$$s(x) = h_x \times d_{\text{truth}} \quad (4)$$

Statements with lower projection values were classified as likely false - this method differs from general classifiers because it directly analyses internal geometry of the model rather than learning a decision boundary (trained on the lies and truths).

E. Causal Truth Injection

Now to check whether truth direction is causally meaningful (compared to correlational) a vector editing intervention was performed on the values. The truth direction was injection into the model’s token embedding weights - Changes in truth scores following this intervention were used as evidence as mechanistic coding.

III. RESULTS

The model successfully learned a latent representation that separates truthful and deceptive statements along a linear direction in the embedding space. Training loss decreased from 4.37 at initialization to 0.017 by step 800, indicating near-perfect convergence between the two classes.

Projection of hidden states onto the learned “truth direction” revealed a clear change:

- Mean Truth Projection: 0.725
- Mean Lie Projection: 0.083

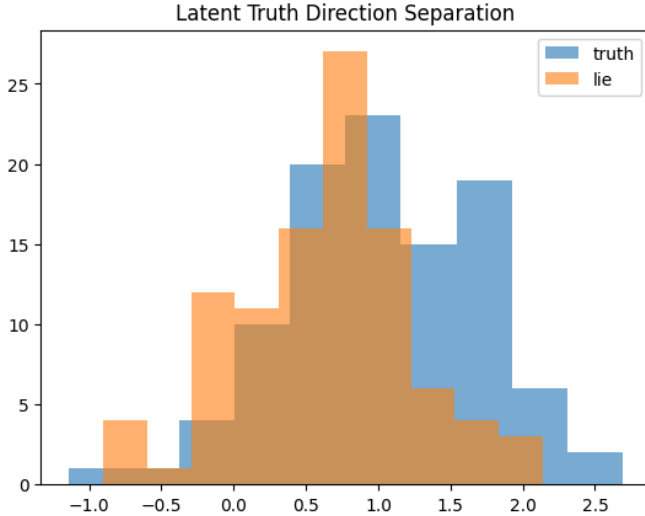


Fig. 2. Histogram of latent truth direction separation for cluster analysis.

The histogram demonstrates that truthful statements cluster at higher projection values, while lies cluster at a lower region of the latent axis. This confirms that truthfulness is encoded as a geometric feature rather than a toggle.

After performing causal intervention by injecting the truth direction into the token embedding layer:

- Truth projection increased to: 1.718
- Lie projection increased to: 1.255

IV. DISCUSSION & CONCLUSION

These findings support the hypothesis that neural networks like transformers encode properties that are semantic such as truthfulness. However, the causal intervention results reveal an important caveat with the original proposition. Injecting the truth direction increased projection scores for both truthful and deceptive statements. This may suggest that extracted direction may not respond to correctness only and could also include linguistic coherence, confidence and other factors.

This study demonstrates that truthfulness can be extracted as a linear latent direction within a neural network trained from scratch. The model forms a geometric representation where truthful statements project more strongly onto a specific axis in embedding space.

Overall, this work contributes a proof-of-concept framework for latent truth detection and causal feature training within neural representations. Now, linking back to the original prerequisite and my overall judgement, I believe that the potential mechanistic interpretability holds, given that in a short time span I was able to gain such insights into latent truthfulness, is extremely great towards AI safety.

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